AquaPlay® Interactive System
Operations & Maintenance Manual
Start-Up Program
VER 2.1

Prepared for:

Project No. 37424

Silliman Aquatic Center

Newark, California

January 27, 2018

Reviewed and submitted by: _____

Tim Kwasnicki P. Eng

www.whitewaterwest.com

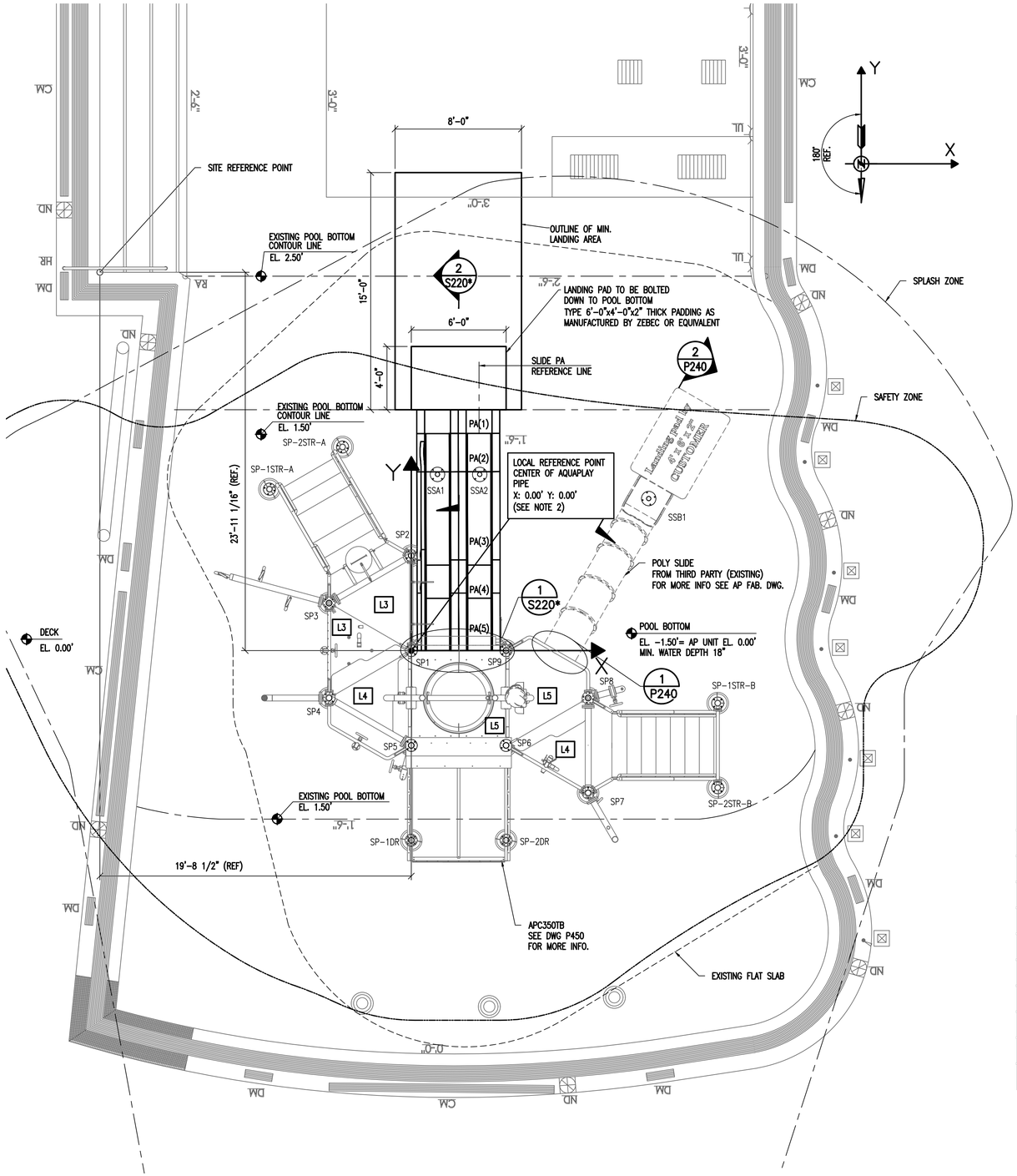
Table of Contents

LOCATION DRAWINGS

Group P Slide Path Layout, Silliman Aquatic Center, Project No.37424 Drawing No. P200 Dated 2017-11-03

SECTION 1. OPERATIONS 5
INTRODUCTION 5
1.1 SIGNAGE 5
1.3 ATTENDANTS 6
1.4 OPERATIONAL RESTRICTIONS DUE TO WEATHER 6
SECTION 2.INTEGRAL WATERSLIDE OPERATIONS 8
MINI MULTI (MM) KIDS BODY SLIDE INTO POOL – SLIDE “PA” 8
1. POSTED SIGNS 8
1.1 Entry Station: 8
1.2 Splash Pool: 9
2. ATTENDANTS 9
2.1 Attendant / top - 1 attendant per double slide 9
2.2 Attendant / splash pool: 1 lifeguard per up to 4 lanes. 9
3. VEHICLES 10
4. WATER FLOW 10
POLY (POLY) KIDS INTO POOL – SLIDE “POLY” 11
1. POSTED SIGNS 11
1.1 Entry Station 11
1.2 Spash Pool 11
2. ATTENDANTS 11
2.1 Attendant / top - 1 attendant per slide 12
2.2 Attendant / splash pool: 1 lifeguard per up to 4 slides. 12
3. VEHICLES 13
4. WATER FLOW 13
SECTION 3. MAINTENANCE 15
3.1 DAILY MAINTENANCE 15
3.2 WEEKLY MAINTENANCE 17
3.3 SEMI-ANNUAL MAINTENANCE 18
3.4 END OF SEASON MAINTENANCE 19
3.5 PAINTING AND REPAIR OF GALVANIZED SURFACES 21
3.6 MAINTENANCE GUIDELINES FOR FRP FLUME 22
SECTION 4. PLAY SEASON START-UP INSTRUCTIONS 28
4.1. BEFORE THE PUMPS ARE STARTED 28
4.2. START-UP 29

4.3. EFFECTS ADJUSTMENT 29
4.4. PLAY TESTING 31



SLIDE PATH LAYOUT

SCALE 1/4"=1'-0"

SLIDE PA CO-ORDINATE TABLE

SECTION NUMBER	PART CODE	DESCRIPTION	ORTN	XCOORD (ft)	YCOORD (ft)	AZIMUTH (deg)	ELEVATION (ft)	GRADE (%)
PA(1)	MMPE-2S-ASSY**	MM Pool Entry - 2 Lane Assembly		4.288	13.730	270.12	1.329	0.00
	MMPE	Mini Multi Pool Entry						
	MMPE	Mini Multi Pool Entry						
PA(2)	MM242STR-2S-ASSY**	MM 738mm STR - 2 Lane Assembly		4.293	11.310	270.12	1.329	0.00
	MM242STR	Mini Multi Straight 242						
	MM242STR	Mini Multi Straight 242						
	MLESTRR	Straight End RH						
PA(3)	MM36CC-2S-ASSY**	MM (3600R 30.00") CC - 2 Lane Assembly		4.306	5.405	270.12	2.911	57.74
	MM36CC	Mini Multi (3600R 30.00") CC						
	MM36CC	Mini Multi (3600R 30.00") CC						
	MLR36CC	ML 3.6m CC Riser						
PA(4)	MM242STR-2S-ASSY**	MM 738mm STR - 2 Lane Assembly		4.310	3.309	270.12	4.121	57.74
	MM242STR	Mini Multi Straight 242						
	MM242STR	Mini Multi Straight 242						
	MLR36STR-SP[738]	Straight Riser 3600 [738mm]						
PA(5)	MM2CV-2S-ASSY**	MM (2000R 30.29") CX - 2 Lane Assembly		4.318	0.005	270.12	5.000	-0.35
	MM2CV	Mini Multi (2000R 30.29") CX						
	MM2CV	Mini Multi (2000R 30.29") CX						
	MLR2CV-RH-END	CX Riser RH END						

Total ride length: 15.990 (ft)

Total number of parts: 14

Total MM ride length: 15.990 (ft)

ENG BOM Version: 2017.10.24 14:22:43 / 0

BOM Notes: Added risers along one side of slide for clearance issue between slide and AP unit

Slide Information details

Silliman Aquatic Center, Newark, CA, Project # 37424

	1	2
Ride Serial Number	37424-PA	37424-Poly
Date of Manufacture	2018	2018
Ride Name	Mini Multi	Poly slide
Product Family	MM	Poly
Ride Speed	maximum 15.0 feet per second	maximum 15.0 feet per second
Water Flow	design 100 US gallons per minute	incidental sprays
Passenger Capacity by Weight	200 pounds per lane	200 pounds
Passenger Capacity by Number	one person per lane	one person
Minimum Rider Height	36 inches	36 inches
Vehicle Type	no vehicles	no vehicles
Patent Number	www.whitewaterwest.com	www.whitewaterwest.com

NOTES:

1. FOR GENERAL NOTES AND SPECIFICATIONS SEE DWG 130.
2. FOR STANDPIPE & SLIDE SUPPORT LOCATIONS SEE DWG P310.

LEGEND:

INDICATES COLUMN LOCATION

REV	REVISION DESCRIPTION/ISSUE RECORD	DATE	PE
0	ISSUED FOR CONSTRUCTION	2017-11-03	

DESIGNED BY MK	DATE 2017-10-18	PROJ. ENG. CHECK	DATE
DRAWN BY YW	DATE 2017-10-18	DRAFTING CHECK	DATE
SIZE D	SCALE PER VIEW	ARCHIVE FILE No. 37424P200	

NOTE:- IMPERIAL DIMENSIONS
ALL ELEVATIONS AND CO-ORDINATES ARE IN FEET
ALL OTHER DIMENSIONS ARE IN FEET AND INCHES

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GROUP P
SLIDE LAYOUT

SILLIMAN AQUATIC CENTER
NEWARK, CA, UNITED STATES

PROJECT No.	DRAWING No.	REVISION
37424	P200	0

PLOT DATE: 2017-11-03 PLOT TIME: 18:52 FILE NAME: 37424P200
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TEMPLATE VERSION: 2017-06-MP

SECTION 1
OPERATIONS

SECTION 1. OPERATIONS

INTRODUCTION

The following guidelines have been provided to assist owners and operators of the Whitewater Interactive AquaPlay ® System with general operations, maintenance and repair information. The procedures described are only recommendations, but if followed will help to ensure trouble free operation of the play equipment.

These guidelines are organized into four sections. Section one contains information that may be used to develop a basic operation program. Section two contains information specific to the operation of the slides on the AquaPlay® structure. Section three contains maintenance and inspection information. Section four outlines start-up procedures that should be followed at the beginning of each operating season to insure that all effects are working and properly balanced.

1.1 Signage

A play element of the size and complexity of a Whitewater Interactive AquaPlay® System generates a lot of activity and excitement among users. We recommend that lifeguards or attendants strongly enforce these rules and encourage parents to do enforce them as well.

Adequate signage is among the most important tools for insuring safe operation of any Whitewater Interactive AquaPlay ® System play unit. Signage reinforces proper behavior among play participants and will aid attendants in promoting safe play. We strongly recommend the installation of at least one sign that is easily observable by children and adults when entering the water play area. While this sign should contain all the information typically posted in pools and waterpark areas, we also recommend that the following items be specifically included on it.

1. **AN ADULT MUST SUPERVISE CHILDREN UNDER 6**.
2. **NO RUNNING** ON OR AROUND THE WATER PLAY UNIT.
3. **NO HEADFIRST SLIDING** ON SLIDES OR IN TUNNELS.
4. **NO CLIMBING** ON PIPES OR HANDRAILS.
5. **NO HANGING OR SWINGING ON PULL ROPES.**
6. VALVES AND HANDLES ARE ACCESSIBLE TO ALLOW CHILDREN AND ADULTS TO THROTTLE AND PLAY WITH THE WATER. WE ENCOURAGE YOU AND YOUR CHILDREN TO TURN, PULL AND OPEN THESE VALVES TO HAVE FUN WITH THE WATER EFFECT YOU CREATE.

1.3 Attendants

The Whitewater Interactive AquaPlay ® System is to be supervised by at least one or more attendants. The attendants are responsible for the safety and orderly conduct of customers, general crowd control and emergency procedures

During operating hours, there should be at least one person on duty at all times who has completed a Red Cross or equivalent Standard First Aid and Safety course. This person should also be competent in carrying out any emergency procedures particular to the unit they are supervising.

1.4 Operational Restrictions Due To Weather

Wind - AquaPlay® structures are designed to withstand wind loads of 80 mph (128 kph). Whitewater requires suspending operation during winds in excess of 40 mph (64 kph).

Lightning- AquaPlay® units incorporate elevated platforms and roof structures. Operation of AP units should be suspended during any storm or weather conditions where lightning may occur.

Rain - Rain will have no effect on the operation of the AP units, with one exception. All flumes rely on the cushioning and braking action of the specified water flow. Rain water can create a very slippery surface and without the braking action of the larger water flow can create an extremely hazardous condition. Under no circumstances should the slides be operated without the main water supply running at specified capacity.

SECTION 2
INTEGRAL WATERSLIDE OPERATIONS

SECTION 2.INTEGRAL WATERSLIDE OPERATIONS

MINI MULTI (MM) kids body slide into Pool – Slide "PA"

The primary operating concern in Multi Lane Body Slides into pool is the possibility of one riding customer colliding with another customer, either in the flume or in the splashdown pool. This type of occurrence can be minimized by the use of well-trained attendants controlling entry and exit conditions. Other safety considerations include: slips and falls in the entry and exit areas, abrasion from the sidewalls, and impact with the splashdown pool surface.

1. Posted Signs

1.1 Entry Station:

Signs should be posted at the entry station of the ride to convey the following rules and regulations:

1. Maximum operational load: 1 person, 200 lbs (90.9 kg), each lane.
2. Children under 36 inches (0.91 m) tall are not permitted on this ride.
3. Children under 6 years of age must be supervised by a responsible person over the age of 16 years to use this ride.
4. Eyeglasses must be securely affixed to riders with head straps.
5. Swim wear with exposed zippers, buckles, rivets or metal ornamentation are not permitted.
6. Only one rider per lane to enter the flume at a time. Single Riders only!
7. Never form chains.
8. Slide must be ridden feet first lying on your back or in a sitting position (sit up to go slower, lie down to go faster).
9. Riders must wait for the attendants start signal before starting the ride.
10. Keep arms and hands inside the flumes at all times.
11. Do not run, dive, stand, kneel, rotate or stop in the slide.
12. At the end of the slide, obey all instructions by splash pool attendant and exit quickly.
13. CAUTION: For safety reasons, pregnant women and persons with heart conditions or back trouble should not ride this slide.

1.2 Splash Pool:

Signs should be posted at the splash pool of the ride to convey the following rules and regulations:

1. Do not block the end of the slide.
2. Leave the splash pool quickly and orderly.
3. No swimming or flotation devices allowed in the splash pool.

2. Attendants

Attendants should be alert to controlling crowd behavior, ensuring safe and orderly entry and exit and controlling the proper entry rate into the slide.

2.1 Attendant / top - 1 attendant per double slide

This requirement may be relaxed to one attendant per level, provided they are able to fulfill all the duties as listed.

Duties:

1. Ensure the crowd behaves in a safe and orderly manner.
2. Ensure all riders meet requirements.
3. Ensure eyeglasses are securely affixed to the riders.
4. Observe bathing suits for zippers, buckles, rivets, etc. and do not allow them on the slide.
5. Observe riders for any physical impairments that compromise safety and do not allow them on the slide.
6. Ensure all riders are in proper riding position before dispatching.
7. The attendant shall dispatch riders one at a time per lane and shall not dispatch a rider until the slide landing area is clear of obstruction and any previous riders.
8. Monitor riders in areas visible from assigned positions.

2.2 Attendant / splash pool: 1 lifeguard per up to 4 lanes.

Duties:

1. Ensure customers conduct themselves in a safe and orderly fashion.
2. Do not allow riders to block the end of a slide. Delay dispatch of next rider if the slide landing area is obstructed.
3. Ensure that all riders exit the pool immediately upon splashdown. Do not permit guests to sit on pool steps or congregate around pool edge.
4. Be prepared to enter splash pool to assist customer out of pool.

5. Do not allow any swimming or flotation devices in splash pool.

NOTE: Special attention is required at the splash pool due to reduced visibility from flowing water and the splash caused by riders entering the pool. The clarity of quiescent splash pool water is normally acceptable if no turbidity is noted and the bottom detail is clearly visible.

3. Vehicles

No vehicles are used in this slide. Patrons ride the slide in bathing suit attire according to instructions above.

4. Water Flow

The water flow should be maintained at 50 U.S. gallons per minute (3.2 litres/second) approximately, per lane, unless directed by Whitewater representative.

POLY (poly) kids into Pool – slide "Poly"

The primary operating concern in the slide is the possibility of one riding customer colliding with another customer, either in the flume or at the end of the slide. This type of occurrence can be minimized by the use of well-trained attendants controlling entry and exit conditions. Other safety considerations include: slips and falls in the entry and exit areas, abrasion from the sidewalls, and impact with the pool surface or bottom of the pool.

1. Posted Signs

1.1 Entry Station

Signs should be posted at the entry station of the ride to convey the following rules and regulations:

1. Maximum operational load 1 person, 200 lbs (90.9 kg).
2. Person under 36 inches tall (0.91 m) are not permitted on this ride. Children under 6 years of age must be closely supervised by an adult.
3. Eyeglasses must be securely affixed to riders with head straps.
4. Swim wear with exposed zippers, buckles, rivets or metal ornamentation are not permitted.
5. Only one rider to enter the flume at a time. Single Riders only!
6. Never form chains.
7. Slide must be ridden feet first lying on your back.
8. Riders must wait for the attendants start signal before starting the ride.
9. Do not run, dive, stand, kneel, rotate or stop in the slide.
10. At the end of the slide, obey all instructions by attendant and exit slide landing area quickly.
11. CAUTION: For safety reasons, pregnant women and persons with heart conditions or back trouble should not ride this slide.

1.2 Spash Pool

Signs should be posted at the splash pool of the ride to convey the following rules and regulations:

1. Do not block the end of the slide.
2. Leave the slide landing area quickly and orderly.

2. Attendants

Attendants should be alert to controlling crowd behavior, ensuring safe and orderly entry and exit and controlling the proper entry rate into the slide.

2.1 Attendant / top - 1 attendant per slide

This requirement may be relaxed to not less than one attendant per platform area or level, provided that the operator ensures that attendants are able to fulfill all duties as listed.

Duties:

1. Ask each rider if they know how to ride the ride.
2. Assure the rider is in the correct body position when ready.
3. Control dispatch of riders one at a time and shall not dispatch a rider until the flume is clear of obstruction and any previous rider.
4. Ensure all riders meet requirements.
5. Ensure eyeglasses are securely affixed to the riders.
6. Observe bathing suits for zippers, buckles, rivets, etc. and do not allow them on the slide.
7. Observe riders for any physical impairments that compromise safety and do not allow them on the slide.
8. Ensure the crowd behaves in a safe and orderly manner.
9. Monitor riders in areas visible from assigned positions.
10. Do not allow riders to block the end of a slide. Delay dispatch of next rider if the slide landing area is obstructed.
11. Ensure that all riders exit the slide landing area immediately upon splashdown. Do not permit guests to congregate around the slide landing area.
12. Be prepared to enter the pool to assist customer.

2.2 Attendant / splash pool: 1 lifeguard per up to 4 slides.

Duties:

1. Ensure customers conduct themselves in a safe and orderly fashion.
2. Do not allow riders to block the end of a slide. Delay dispatch of next rider if the slide landing area is obstructed.
3. Be prepared to enter splash pool to assist customer out of pool.

   NOTE: Special attention is required at the splash pool due to reduced visibility from flowing water and the splash caused by riders entering the pool. The clarity of quiescent splash pool water is normally acceptable if no turbidity is noted and the bottom detail is clearly visible.

3. Vehicles

No vehicles are used in this slide. Patrons ride the slide in bathing suit attire according to instructions above.

4. Water Flow

There is no dedicated water flow, rather it is wetted from splash and from patrons.

SECTION 3
MAINTENANCE

SECTION 3. MAINTENANCE

Regular maintenance of the play unit is important not just for appearance or function, but in keeping the play unit *safe*. Although you will probably have a maintenance staff to perform repair work on the structure, attendants must also play a part through observing the play structure during morning start-up and daily operation. The attendants will be the first line of defense in spotting potential hazards that might develop. The following text outlines maintenance and inspection schedules as well as some general repair information.

MAINTENANCE AND INSPECTION SCHEDULES

Daily	* Check operating pressure and flow to all effects.
	* Visually Inspect climbing and safety nets
	* Visually Inspect web crawl tunnels
Weekly	* Visually Inspect large tipping bucket assembly.
	* Visually Inspect swinging bridge joints.
	* Inspect waterslides and/or tunnel slides.
	* Check operation of jets/ nozzles and clean as required.
	* Check valve operating assemblies.
End of Season	* Drain play structure.
	* Inspect entire structure. Repair or replace components as needed.
	* Clean components as needed.
	* Inspect and repair any damaged paint and/or galvanizing.
	* Remove and store small parts.
Semi-Annual	* Tipping Bucket Inspection

The following sections will describe the elements of the above listed maintenance procedures in detail.

3.1 Daily Maintenance

3.1.1 Operating Pressures:

Each water line that runs to a Whitewater Interactive AquaPlay® ® ® System play structure is typically valved either in the pump room or at the structure itself. These valves are for the use of the owner/operator to regulate and adjust the flow of water to each individual effect to achieve safe operating pressures. Once initial adjustments are made at first start-up, the valves should be **marked or tagged** so that they can be returned to the desired position if they are accidentally moved or

when the play unit is started-up at the beginning of a new season. All of these controls must be in a location that is not accessible to the general public.

Insuring that these initial adjustments are properly maintained is important for safe and fun use of the play structure. The valve settings should be visually checked or the play unit should be observed in operation at the beginning of the day to insure that none of the correct settings have been altered.

3.1.2 Nets:

Inspecting and maintaining climbing nets is vitally important for safe operation of the play structure. A daily inspection of these components should include the following:

* Visually check all net and or steel cables for wear and tear.
* Visually Check that net lashings are securely tied and in good condition.
* Visually check the climbing and safety nets for cuts, tears or other damage.
* Check for excessive sagging. The nets will stretch over time and will require periodic adjustment.

3.1.3. Web Crawl Tunnels:

As with the climbing nets, the web crawl tunnels should be inspected daily:

* Check webbing for cuts, tears or other damage.
* Check that lashings are securely tied and in good condition.

3.1.4 Slide Inspections:

If Integrated Slides are present daily inspections should be made to check for:

* Obstructions in the slide path
* Cracks or chips in the sliding surface
* Rough patchwork at joints or cracks
* Excessive movement of the flume when walked on
* Inspect slides and/or tunnel slides for cracks, breaks or tears.
* Inspect slide section joints. Check that all bolts are in place and securely fastened.
* Check slide supports. Make sure all fasteners are in place and secure.

3.1.5 Pool Chemical Inspections:

Proper water chemical balance should be obtained each day before the facility is opened to the public. While the facility is operational water quality testing should be performed at intervals in accordance with local health authorities. Inspections should also be made during periods of heaviest usage to ensure that the circulation

and filtration systems are handling peak loads. The operator should keep logs of all water testing performed.

The operator must be sure that customers cannot come in contact with the water treatment, re-circulation and filtration equipment. Mechanical and chemical storage rooms should be locked at all times to prevent customer access.

3.2 Weekly Maintenance

3.2.1. Large Tipping Bucket

* Visually inspect the torque indicators on the lock nuts retaining the bucket bearings.
* The torque indicator is a paint line drawn across both lock nuts. The mark on both nuts must be in alignment.
* The bucket bearing lock nuts retaining the bucket bearings, and the torque indicator marks, can be readily observed on the underside of the bucket bearing support and can be seen from the platform below. If desired, they can be viewed through a set of binoculars for greater clarity.
* If the torque indicator marks are out of alignment and show any relative rotational change, then it indicates that the lock nuts have rotated and have lost their correct torque. A closer inspection of the bucket bearings per the Semi-annual Maintenance is required immediately.
* If the bucket has a foam core facade, visually check facade for delamination or any other excessive wear and tear.
* If the bucket has a wood plank facade, visually check that planking is securely fastened and in good condition.

3.2.2 Swinging Wood Bridge

* Inspect the safety net handrails as for net climbs (above).
* Check support bars and attachments.

3.2.3 Jets and Nozzles:

All play structure jets and nozzles should be routinely checked for proper operation. Many of the piping effects consist simply of nipples recessed in the piping. These nipples will sometimes become clogged with debris. Usually, inserting a bottlebrush or a piece of wire into the pipe orifice will suffice to clean out the debris.

There are several overhead nozzles that should also be checked. Specifically, these consist of the umbrella jet and the pull rope valve assemblies. Like the pipe jets, these nozzles will sometimes become clogged with debris. When this occurs, the nozzles should be removed and cleaned.

The umbrella jet is a two piece assembly consisting of a 3" brass sleeve and a brass cone that can be adjusted to allow more or less water to flow through the sleeve. It is the gap between the sleeve and the cone that will become clogged. To disassemble the umbrella jet, simply remove the hex screw from the top of the cone, then turn the cone counter clockwise until it comes off.

There are two problems that can occur with the pull rope valves. Like the other nozzles, they will sometimes become clogged. Also, these valves will occasionally freeze up, causing the valve to remain in an open position when the rope is pulled. Both of these conditions can be corrected by disassembling the valve and cleaning the valve components.

To disassemble a pull rope valve, remove the threaded cap from the back of the valve with a wrench. Once the cap is removed, a spring will be visible. Remove the spring and then push the valve stem inward to remove the operating mechanism. Clean all parts with soap and water or a mild solvent and allow to dry. To correct a sticking valve, remove the small O-ring from the valve stem and clean the stem and stem sleeve with emery cloth. Then simply replace the O-ring and reassemble the valve.

3.2.4 Valve Assemblies:

The final item of weekly maintenance is to check the participatory valve assemblies. These consist of the handwheels and lever handles. Check that handwheels and lever handles are securely bolted to the valve bodies. Check the resilient handwheels for damage. Open and close the valves to check that they are working properly. Check throttle valves to insure that the protective caps are in place.

3.3 Semi-Annual Maintenance

3.3.1 Large Tipping Bucket:

* Visually check that the bucket support frame is secure to the play structure stand pipes.
* Visually inspect the bucket pivot shaft for signs of fatigue (cracking of metal or welds). Some deflection (sagging) of the pivot shaft is normal.
* Make sure that bearings are secure and in good condition. Check that bucket swings freely.
* If the bucket has a foam core facade, check facade for delamination or any other excessive wear and tear.
* If the bucket has a wood plank facade, check that planking is securely fastened and in good condition.

3.4 End Of Season Maintenance

The end of the operating season is the time when a thorough inspection of the entire play structure should be performed. This ensures that if any repairs are needed, they can be performed before the next operating season.

3.4.1 Drain Play Structure:

The first item of maintenance is to drain the pool and play unit

of all water. The majority of water will drain out when the below grade plumbing system is drained. Following is a checklist of items that should be done at the unit itself to insure proper drainage.

* Drain the large tipping bucket. There is a ball valve or threaded plug at the base of the bucket for draining. Leave the drain open for the winter season.

* Open all handwheel and lever operated valves. Leave these valves open for the winter season. Make sure all throttle valves on the unit are open.

* Make sure all pull rope valves are open during draining.

* Make sure all basins, water curtains and small tipping buckets are empty. Make sure all hose- bib basin feed valves are open.

* After draining basins, operate one-man and two-man pumps to remove any remaining water.

* Access the manifolds underneath the platforms. The manifold has been equipped with a drain plug. Remove the plug and drain the manifold. Make sure no water traps are present in the above grade plumbing from the manifold. If necessary, tap and drain the plumbing to remove water.

* Operate the bucket conveyor to insure all buckets are empty.

* Operate waterguns to allow any water to drain out.

* Make sure all waterwheels, runnels, etc. are emptied of standing water.

3.4.2 Inspect all Components:

The entire play structure should be thoroughly examined before the winter season for any damage to components. This inspection should include all items previously mentioned in addition to the following:

* Inspect all flanged connections. Check that all bolts are in place and securely tightened.

* Inspect all roof structures. Wood roofs should be checked to insure that all structural supports are securely bolted. Wood members should be inspected for rot or other structural damage. Check that all planking is securely nailed. Sheet metal roofing should be inspected to insure that metal is securely fastened. Check that

any roof facades are secure. Canvas roof poles should be checked to see that keepers are in place and bolted. Check canvas for tears or fraying.

* Inspect peripheral play equipment such as the pools and falls, spinning tray, etc.. Make sure equipment is working properly and all parts are in place. Make sure all bolted connections are secure. Grease spinning tray gear box as needed.

* Inspect fiberglass platforms and stairs for cracks or excessive wear to the integral non-slip surfaces. Check platform frames to see that all fasteners are in place and tightened.

* Check that all handrails are securely bolted.

* Check all pull ropes for excessive wear.

3.4.3 Clean Components:

To insure long life and to maintain the visual appearance of your play structure it is very important that all painted surfaces are cleaned and checked for any damage.

Maintenance requires removal of surface dirt, grease and suntan oil by cleaning the flumes with a soft-bristled brush and mild detergent such as 3M - “Multi Purpose Boat Soap”. For hard water buildup (a white chalky discoloration) removal/maintenance use “C.L.R.” or other cleaners specific for this purpose. For removal of difficult stains use a cut polish or rubbing compound and finish wax or a combination product such as 3M “Marine One-Step Fiberglass Restorer and Wax”.

Use only soft brushes. Wire brushes or abrasive scrubbers will damage the painted surfaces.

3.4.4 Painted Surfaces Inspection:

Each play structure is galvanized inside and out as primary protection then is primed and painted with two coats of urethane paint as secondary protection. If the paint has been chipped or removed, these areas must be carefully inspected to see if the galvanizing has been damaged, if so, you will need to remove the paint in the immediate area, prime and repaint.

On areas where the paint has only been chipped, and the galvanizing has not been disturbed, simple touch-up will protect from additional damage. In some applications Zinc Anodes may be provided. They function through Sacrificial Corrosion and must be left as bare metal. They should be replaced once they have deteriorated too much.

Your play structure was originally supplied with touch-up paint in all colors. If additional coating materials are needed you may contact Whitewater Interactive AquaPlay® ® ® Systems Richmond to obtain the necessary color code information for your play structure. More specific instructions for surface preparation, paint mixing and application are included in this document, see Section 3.4

3.4.5 Remove Small Parts:

Some small parts can be removed during the off season and stored for protection from corrosion and ultraviolet exposure. These include handwheels and lever handles, waterguns, hose jets and small nozzles such as pull rope valves and umbrella jets. These parts are relatively easy to remove and do not require large amounts of space for storage. Each part should be clearly tagged to facilitate easy replacement at start up of the operating season. For waterguns, it is recommended that they be blown out with air to remove any debris that might clog the nozzles. If nozzles and jets are removed, the receptacles should be plugged until the jets are re-installed.

If winter conditions apply – Palm head with fronds are to be stored indoors. Show, ice, sleet and freezing temperature can damage the palm leaves. The trunk can remain outdoors.

3.5 Painting And Repair Of Galvanized Surfaces

The painted surfaces of the play structure consists of a layer of galvanizing followed by an epoxy primer and an over coating of polyurethane enamel paint. When touch-up work is required, there are several important items to remember:

3.5.1. Take Care To Protect The Galvanizing.

The galvanizing process acts as the primary protective coating for the steel piping, handrails and hardware. If the galvanized surfaces are damaged or removed, rust problems are more likely to develop. When sanding surfaces in preparation for painting, try to sand only the paint, leaving the galvanized surface intact.

If the galvanizing has been damaged prior to touch-up and rust has developed, it is important to sand down to bare metal prior to painting. The epoxy primer will act as a seal over the bare steel. If the rust is heavy and difficult to remove, a rust preventer / inhibitor can be applied prior to painting. This inhibitor reacts chemically with the rust, preventing further oxidation. Repair to damaged galvanized surfaces can also be made with various Zinc Rich Primers known as cold Galvanizing.

3.5.2. Make Sure All Surfaces To Be Painted Are Clean and Dry.

In locations with water hardness problems, there may be a build-up that needs to be removed prior to the surface preparation. This scale may be removed with an application of a weak solution of muriatic or phosphoric Acid. Scrub the surface with soft brushes only, so as not to damage the paint or galvanizing and rinse with clean water.

Particles of dirt, oil, water or other substances adhering to the pipe during painting will cause both application and adhesion problems resulting in a poor quality finish. After preparing the surfaces for paint, wipe surfaces down with a tack cloth (supplied

in the detail kit). This will remove the fine particles from the surface leaving it clean to paint.

3.5.3. Mix Only Small Quantities Of Paint At a Time.

The primers and paints used by White Water are activated coatings, and therefore must be used quickly once mixed. After a period of time the paints and primers will become hard and unusable (see paint instructions in the detail kit supplied with every job).Mix only quantities which can be used in 1 hour.

3.5.4. Be Sure Surfaces Are Warm.

Whitewater primers and paints may be applied at a temperature range between 50 and 85 degrees F (10 and 29 degrees C), with 70 degrees F (21 C) being ideal. For small areas of spot touch-up, paints may be applied at colder temperatures, although it is not generally recommended.

Spraying is the ideal method for applying Whitewater coatings. However the detail kit contains all primers paints catalysts brushes and mixing containers for spot paint repair.

3.5.5. Preparation and Application

Sand entire area to be painted with 120 grit sandpaper (supplied), feathering out all edges. After sanding, wipe down with tack cloth (supplied).

Apply 2 coats of primer allowing primer to set between coats (tacky to touch)

Apply 2 coats of paint. Allow paint to dry to tack between coats (10 to 20 minutes). Allow paint to cure a minimum of 36 hours before immersion in water.

IF YOU HAVE QUESTIONS ABOUT THESE INSTRUCTIONS OR NEED FURTHER HELP IN REGARDS TO PLAY SYSTEM MAINTENANCE, PLEASE CALL Whitewater Interactive AquaPlay® ® ® Systems AT (604) 273-1068 AND ASK FOR TECHNICAL ASSISTANCE.

3.6 Maintenance Guidelines For Frp Flume

Properly maintained, your water slide will give you years of service and keep sliding the way it was meant to be - SAFE and FUN. By keeping the slide and its components properly maintained you will be ensuring high performance and long life.

3.6.1 Inspections of Flume:

NOTE: Before performing any inspection or maintenance work on the waterslide and related equipment, ensure that all equipment is electrically isolated using recognized equipment procedures. Equipment should not be turned on until all personnel are clear from the ride and accounted for.

Each slide should be walked through daily before start-up to check for:

- obstructions in the slide path
- cracks or chips in the sliding surface
- rough patchwork
- leaking seals at the joints
- loose risers on the turns
- bubble in slide path

3.6.2 Cleaning:

Maintenance requires removal of surface dirt, grease and suntan oil by cleaning the flumes with a soft-bristled brush and mild detergent such as 3M - “Multi Purpose Boat Soap”. For hard water buildup (a white chalky discoloration) removal/maintenance use “C.L.R.” or other cleaners specific for this purpose. For removal of difficult stains use a cut polish or rubbing compound and finish wax or a combination product such as 3M “Marine One-Step Fiberglass Restorer and Wax”.

3.6.3 Waxing:

At least twice each year the fiberglass finish should be waxed and buffed with high quality paste wax or polish. On scratched, dull or faded areas use a fine buffing compound to restore the original gloss before waxing. Prolonged exposure to the ultra-violet rays of the sun can, in time, cause discoloring and fading. We recommend using paste waxes or polishes containing ultra-violet screens.

CAUTION: If you use power tools for buffing or polishing, use extreme care to prevent overheating the surface. Overheating a rubbing compound burns small gritty particles into the slide surface causing blistering and streaking. This may require the area to be sanded and repainted.

3.6.4 Joints:

Slide joints play an important role in rider comfort and pleasure. Proper maintenance will keep the joints smooth and trouble free for years.

3.6.4.1 Joint Caulking: All fiberglass sections will expand and contract with temperature changes. Therefore, the joints between sections are filled with a flexible, elastic caulking. Should the joints need re-caulking, all the old filler should be removed, the fiberglass cleaned and the joint re-filled using *Sikaflex 1a Polyurethane Construction Sealant*. DO NOT use a plastic filler that will harden. Under no circumstances should the joints be fiberglassed over. Excess joint caulking should be removed from the slide surface by scraping it off with a putty knife. Any remaining traces should be wiped off with lacquer thinner applied to a

rag. Do not pour lacquer thinner directly onto a flume joint as it can dissolve the sealant.

3.6.4.2 Joint Roughness: Joints may become rough with use, especially if metal articles are allowed down the slide. For minor chips and cracks, the joints may be fixed using the procedures outlined in Section 5.

More serious cases may require power sanding that will cut through the gelcoat. This will require a spray application of new gelcoat by a trained person, finished by sanding and polishing and waxing as outlined in Section 5.

3.6.5 Patching:

Chips and cracks may appear in the fiberglass flumes from normal usage. The following sections outline procedures to repair minor damage to the slides. However, due to the hazardous nature of the materials used and the variability of application methods, we recommend that a qualified fiberglass laminator perform the repairs.

3.6.5.1 Minor Chips and Cracks: In areas that have chips or cracks in the gelcoat only, or have a scratch that is deep enough to penetrate through the gelcoat to the fiberglass but not deep enough to go completely through the laminate, follow the procedures given in Section 5.4.

NOTE: For damage that penetrates completely through or deep into the laminate, we recommend that a qualified fiberglass laminator perform the repairs.

3.6.5.2 Surface Cracks: Hairline cracks, sometimes called spider webbing or star cracks, may develop in the gelcoat or surface coating of the fiberglass product. This is caused by a combination of weathering, vibrations, and/or impacts. Although unsightly, they do not affect structural strength. The hairline cracks can be fixed by sanding out the affected area with 100 grit sandpaper and re-coating the surface with gelcoat. Follow the patching instructions in Section 5.4 omitting steps 4 through 6.

3.6.5.3 Cracks Around Flanges: The area around the flanges of the flumes contains a thick layer of gelcoat which is very stiff. Stresses from normal use may cause cracks in the gelcoat around the flanges. These are not structural cracks. Flange cracks may be remedied by sanding the affected area with 100 grit sandpaper and re-coating with gelcoat. Follow the patching instructions in Section 5.4 omitting steps 4 through 6.

3.6.5.4 Repair Procedures:

********** W A R N I N G**********

Patching kits contain toxic, combustible and corrosive elements. Use only under well ventilated conditions. Do not inhale fumes. Harmful if swallowed. If contents are swallowed or inhaled, seek medical attention regarding polyester resins. If

contacted to eyes, flush with water for 15 minutes and seek medical attention. If contents come in contact with skin or clothing, clean the area immediately with acetone.

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Before you start, you will need:

a) Sandpaper - 36 - 100 grit; 220 - 320 grit wet and dry; 600 grit wet and dry

b) Sanding block

c) Acetone

d) 2" masking tape

e) Utility knife

f) Putty knife

g) Mixing board

h) Mixing cup

i) Fiberglass resin hardener (catalyst), use a peroxide based catalyst (MEKP)

j) Patching compound

k) Gelcoat paint

l) 2" fine paint brush

m) Automotive cut polish

n) High quality paste wax such as "Turtle" wax

3.6.5.4.1 Storage Instructions: Store uncatalyzed patching compound and gelcoat in closed metal containers in a cool area below 73 degrees Fahrenheit (23 degrees Celsius) out of direct sunlight. Do not allow to freeze. Store all materials in their original containers according to manufacturer's instructions.

3.6.5.4.2 Mixing Instructions: Mix only small amounts at one time. The mixtures will harden in 20 minutes to one (1) hour, depending on the temperature and the amount of catalyst added. The mixture will harden faster at higher temperatures and with increasing amounts of catalyst added. Use caution when adding catalyst, over catalyzing will result in the mixture cracking. Do not pour catalyzed mixtures back into the can. Do not reduce materials with a "conventional" paint or lacquer thinner.

Patching Compound: add one (1) part catalyst to 100 parts patching compound on the mixing board. Mix well.

Gelcoat: add one (1) part catalyst to 100 parts gelcoat in the mixing cup. Mix well.

3.6.5.4.3 Patching Instructions:

1. Sand area to be patched with 36 -100 grit sandpaper. Feather sand to moulded surface and remove all rough edges.

2. Wash area clean with acetone and allow to dry.

3. Mask off area to be patched with 2" masking tape.

4. Using a putty knife, spread the patching compound smoothly over the prepared area to the top of the tape.

   NOTE: area must be dry and clear of contaminants, such as dirt and sanded particles, etc.

5. Keep the patched area dry and allow 20 minutes to one (1) hour to harden.

6. After the patch has hardened, sand the patched area flush to the surrounding surface with 100 grit sandpaper.

   NOTE: Sanding will remove the masking tape. Re-tape the area before painting.

7. Using a 2" fine paint brush, paint the area with gelcoat. Allow 20 minutes to one (1) hour to dry.

   NOTE: the paintbrush and tools can be cleaned with acetone.

After the gelcoat has dried, remove the masking tape. Using a sanding block, wet sand with 220 - 320 grit sandpaper. Repeat with 600 grit sandpaper.

Buff painted surface with auto cut polish and wash clean with water.

10. Wax and buff painted surface with a high quality paste wax.

SECTION 4
PLAY SEASON START-UP INSTRUCTIONS

SECTION 4. PLAY SEASON START-UP INSTRUCTIONS

The overall design intent of a Whitewater Interactive AquaPlay® System water play unit is to allow play participants to turn valves, pull ropes and operate various mechanisms to control the flow of water for their fun and enjoyment. However, water pressures must be adjusted to guarantee safe operation. To accomplish this, we have insured that all AquaPlay® effects are adjustable either at the equipment space manifolds or by using adjustable throttle valves on the unit. Once the operator properly sets the highest operating levels, the play participants can open or close all interactive valves to adjust the effects as they wish. They can always throttle the effects down, but never higher than what has been pre-set by the owner/operator.

Initial valve settings will have been made by a Whitewater Interactive AquaPlay® Systems representative and the owner/operator upon the first season start-up of the unit. Only minor adjustments should be required from season to season provided that initial settings have been maintained.

4.1. Before The Pumps Are Started

1. Remove small orifice nozzles from the play structure where possible. This will prevent clogging of the nozzles by debris that may be in the lines. These nozzles include the overhead pull rope valves, umbrella jet and waterguns.
2. Open all valves on the manifolds and all butterfly valves on the structure.
3. Replace any basin and/or manifold plugs that were removed for the winter season. Review the winter draining checklist above as a reference for checking effects prior to start-up.
4. Since the play unit will be active during start-up, the person that is to do any valve adjustment in the pool area should wear appropriate attire. Once initial adjustments are completed, further adjustment should not be required during the operating season unless problems develop.
5. A minimum of three people should be on hand for start-up. One should be an operations person with authority to make decisions regarding visual impact and safe operating pressures for each effect. A second person will be in the pool to perform valve adjustments as needed. A third person should stand by the pumps at all times to turn the system on and off as necessary. A radio should be used to communicate quickly with the pump operator in case problems occur.
6. Be sure that all strainers are clean and valves are open on the pump suction and discharge lines. The pump(s) should be "bumped" prior to start-up to check for proper rotation.

4.2. Start-Up

1. When the play structure is first operated, all personnel should stand at least 20' away from all effects. Some jets may spray out a long distance. This may catch some people unaware and possibly cause injury if they are too close.
2. Turn the pump on. It will take a minute or so for the water to fill the pipes and basins. Continue to run the pumps until all debris and dirt has been cleaned out of the system, then shut down.
3. With the pump off, re-install any jets or nozzles that were removed for the winter season.
4. The pump can now be re-started and allowed to run while any needed adjustments are made.

4.3. Effects Adjustment

The following is a brief description of the most common water effects and how these effects should appear and operate.

4.3.1 Single Angle Jets:

These are single orifice jets and are usually located on a pipe elbow. The jets should be adjusted to shoot as far as possible and stay inside of the pool (preferably about 15 to 20 feet). Be sure that the required pressure to do this is safe.

4.3.2 Overhead Rope Pulls:

There are two different types of rope pull effects. One is a full cone spray (similar to a shower head) that sprays water down when the rope is pulled. The other is simply a straight length of pipe that drops a stream of water when the rope is pulled. Sometimes pull ropes are used as participant showers and sometimes they are used to feed other effects such as spinning trays.

If used as showers, the valve for these should be adjusted so the pressure from the sprays will not hurt your face if you look up into them.

4.3.3 Geyser Jet:

This jet should be adjusted so that it shoots up approximately one to two feet above the platform when opened.

4.3.4 Up Jets:

These jets are designed to shoot streams of water directly up into the air. The jets should project about three to four feet up in the air.

4.3.5 Bar Jets:

These jets shoot streams of water straight down. They should be adjusted so as not to sting the face of anyone looking directly up at them.

4.3.6 Water Curtains:

The objective of adjusting water curtains is to get as thick a sheet of water over the edge as possible without it breaking up while retaining a translucent quality. If the water curtain has too little flow, the water will come over intermittently in sections and will break up a foot or two from the top. If the flow is too great the water will be roughly textured, heavy, and you will not be able to see through it. The correct adjustment will result in a clean, crisp curtain of water about 1" thick all the way to the pool water level.

4.3.7 Angle Jets:

These jets are designed to shoot up and out at a 45° angle, arching into the pool about 8 or 10 feet, over the heads of those in the pool.

4.3.8 Arch Jets:

These jets should be adjusted to produce streams of water which shoot out horizontally four to six feet, then arch down into the pool.

4.3.9 Waterwheels:

Waterwheels must be adjusted so that the wheels rotate in a steady fashion. Too much flow may cause a wheel to spin without filling or dumping. Too little flow and the wheel will not turn.

4.3.10 Large Overhead Tipping Bucket:

The tipping bucket has been designed to dump when filled to a certain level. Therefore, the adjustment is to determine how often the bucket is to dump. The higher the flow to the bucket, the less time between dump cycles.

4.3.11 Waterguns:

The waterguns are one of the most critical items for adjustment. Since the play participants can aim them at others, the pressure must be carefully adjusted so as not to hurt them.

The guns are intentionally placed on the platforms at higher elevations so they will not shoot directly at the body or head. Rather, the water will fall or arch down on the target. Since adults are taller the stream of water is more of a problem for them. The best way to adjust these is to have an adult stand at various locations while someone else "shoots" them with water to determine what is safe.

4.3.12 Finally:

Once you have balanced all the effects, go around and turn all of the valves that are to be operated by the users completely off and on a few times to make sure that each one works correctly and that the maximum flow does not exceed what you wanted.

Once all of the effects have been balanced, turn the pump off and replace the PVC panels around all of the platforms.

MAKE SURE ALL THE PANELS ARE ATTACHED SECURELY TO THE STRUCTURE SO THERE IS NO POSSIBILITY OF CHILDREN REMOVING THEM AND BECOMING ENTRAPPED UNDERNEATH THE PLATFORMS.

4.4. Play Testing

We recommend at least an hour or so of play testing before you open the play structure to the public. This should involve bringing in several park personnel and having them play randomly on the structure for a period of time.

If problems should become apparent that are part of the structure (i.e., having to do with railings, steps, piping, etc.) or are what you perceive as improper or unsafe operation, shut the system down and contact this office immediately. If the problem requires immediate attention, we will have a representative visit the site as soon as possible to correct it.

DO NOT MAKE ANY MODIFICATIONS TO THE UNIT YOURSELF WITHOUT FIRST CONTACTING THE Whitewater AquaPlay® OFFICE. Tel: 604-273-1068.

Once the play structure has been play tested and you are satisfied that it is operating correctly, you may open it to the public. Since most Whitewater Interactive AquaPlay ® structures are fairly complicated, we recommend that you observe all operations carefully the first few days to ensure that all the "bugs" have been worked out and the unit is operating at safe water pressures.

Please contact Whitewater AquaPlay® at (604) 273-1068 and ask for technical assistance if you have any questions or need to discuss maintenance or start-up procedures of your play structure. Please remember, we are at your service to assist with any aspect of operation

Safe water flow

The water flow rate is important to the safe operation of the ride. Operators shall make sure that the water supply is adequate. Insufficient water volume can result in higher speed. The water tends to restrain and cushion motions and impacts as well as reduce speed.

Evacuation and Emergency Procedures

In an emergency situation affecting the slide:

1. Top attendants shall cease dispatch of all riders.
2. If water flow is impeding the remedy of the emergency situation, use E-stop to turn off power to the pump(s).
3. Riders in the slide shall continue down the slide and exit normally if possible.

4. Attendants shall ensure all riders have exited the slide either through the bottom or at egress locations.
5. Patrons may remain queued on tower and stairs if they are not affecting the emergency situation.
6. If necessary to evacuate the tower, patrons shall be guided by top attendant to exit via the stairs from the start tower or egress location.
7. Slide should remain closed until cause of emergency situation is investigated and resolved.
8. In case of serious injuries or problems, operator should notify manufacturer and other regulatory bodies as required by law.

WHITEWATER

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